Trauma-Informed Data Donation: Integrating Expert and Donor Perspectives on Designing Against Re-Traumatization During Collection of Sexual Violence Data

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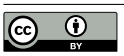
Abstract

Data donation has received attention as a more consensual means of collecting personal data for scientific inquiry and AI technology. Yet the nature of data often donated–such as harmful online messages and menstrual tracking logs–carries risk of retraumatization (the forced reliving of traumatic experience). While the well-being of data donors is considered in prior work, approaches to retraumatization remain ad hoc. We present Trauma-Informed Data Donation (TIDD): a context-specific, exploratory design framework for adapting the Trauma-Informed Approach (TIA) from the Public Health domain to data donation. TIDD was the product of a 2-year research through design process with experts on sexual violence and trauma, and observational interviews of data donors. We use a case study applying TIDD to our custom data donation platform, Ube, as an invitation for designers to consider how TIDD could be used as a malleable foundation for donation of data associated with other forms of trauma.

Keywords

Data Donation, Consent, Trauma, Retraumatization, Sexual Violence

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1 Introduction

Human-centered design has taken a turn towards trauma [9, 34, 35, 71, 75, 105–107, 117, 118], defined as the adverse effect of a personally distressing experience on an individual's physical and emotional well-being [48]. This rise of a trauma orientation in HCI has manifested through the design of technologies intended to address trauma [1, 20, 61, 85, 90, 103, 120], and in the explicit foregrounding of trauma within technology design processes and methods [106, 117, 118], even for technologies that are not necessarily "about" trauma [9, 34, 71, 75]. Incorporating trauma in design practice is most often pursued through adaptation of the Trauma-Informed Approach (TIA)–a framework created by the Substance Abuse and Mental Health Services Administration (SAMHSA) [48] to guide the construction of interactions with, and services for, individuals who have experienced trauma. The crux of TIA is recognition and mitigation of retraumatization; the "reliving [of] stress reactions experienced as a result of a traumatic event when faced with a new, similar incident" [48].

While popularized in public health domains and clinical settings, researchers have sought to translate TIA to the general HCI space through Trauma-Informed Computing principles [23], and to specific application areas like trauma-informed social media [94], in pursuit of mitigating retraumatization incurred through the act of using technology. The HCI literature in recent years exhibits various technologies touted as trauma-informed [52, 81, 91], along with trauma-informed design heuristics [33], and examples of technology design teams deliberately constructed to involve experts on trauma [1, 112, 118] and anticipated users who may have been traumatized [83].

In this paper, we argue for the application of TIA to data donation platforms, which enable the active and deliberate transfer of one's personal data to a third party to support scientific inquiry [38, 73] or improve data-driven technologies (e.g., [3, 5, 8, 86]). Data donation platforms are typically standalone applications that facilitate the uploading of a donor's data that originated from a separate source or application, along with contextualization or labeling of that data (e.g., [7, 36]). Prior data donation efforts have focused on various data sources including wearable devices [38], social media accounts [86], speech records from voice assistants [37], browser histories [66], and medical records [58, 98, 101].

The types of data sought for donation are often "sensitive" [41, 44] and "intimate" [40] in nature, which we argue carries inherent risk of retraumatization due to potential association with traumatic experience. Examples include donation of social platform data representative of harm against teens [86], sexual assault [117, 118], menstrual tracking logs [38] (especially in light of fluctuating legal regulation of women's reproductive cycles [4, 28, 45]), and health data that could be indicative of serious illness [29, 59, 98]. To the credit of prior work, the well-being of donors is often consciously considered in data donation process and interface design [38, 40, 41], yet these approaches are either ad hoc (they are not based in a formal or unifying theory of well-being) or rooted in tangential goals such as supporting informed consent [37, 59, 114].

To address this gap, we present Trauma-Informed Data Donation (TIDD): a malleable, exploratory design framework for adapting the Trauma-Informed Approach (TIA) from the Public Health domain to data donation. We explore a case study through which we created TIDD and applied the trauma-informed approach to design and development of a custom-made data donation platform, Ube, for collecting data about (non)consensual sexual experiences between dating app users. Contributions of the paper include:

(1) A context-specific and modifiable foundation for incorporating the trauma-informed approach into data donation processes called Trauma-Informed Data Donation (TIDD).
(2) A case study of applying TIDD to a data donation app for online dating sexual violence data, Ube, that was designed, developed, and assessed with input from experts on trauma and sexual violence, as well as real data donors. The case study serves to inform or inspire permutations of TIDD for data indicative of other forms of trauma.
(3) Methodological reflections on trauma-informed design, particularly how to evaluate trauma-informed technologies in their adherence to trauma-informed principles.

2 Background

2.1 Trauma in HCI

Trauma has had a growing presence in HCI research and design. While explicit use of the term "trauma" is a relatively new phenomenon in HCI [23], research and design efforts have long put attention toward computer-mediated experiences associated with trauma. This includes study of computer-mediated harms known to be associated with trauma [30, 60, 100, 121], design of new technologies that can address or mitigate trauma [113], and involvement of stakeholders in research and design who may have been traumatized [49, 118].

Approximately 70% of the global population has experienced trauma at least once in their lifetime [12, 68]. While succinct definitions of trauma remain elusive, SAMHSA defines how a person is affected by traumatic experience with the three "E's" [48]: Event, Experience of the event, and Effects. Trauma is instigated by an Event that the body senses as dangerous, such as natural disasters, violence, or life-threatening neglect [48, 88]. How a person Experiences an event, meaning how one "labels, assigns meaning to, and is disrupted physically and physiologically by an event" ([48], p. 8), decides whether a person will experience it as traumatic or not. Third is the long-lasting Effects of the event [48], which can occur immediately or have a delayed onset. One noteworthy long-lasting effect of trauma that warrants special consideration in research practices is the risk of retraumatization, whereby reflecting on a traumatic event can trigger a person's experiences of trauma to resurface, and lead to negative mental health effects [23, 47, 48, 69, 110].

2.1.1 The Trauma-Informed Approach. As retraumatization and other adverse effects of trauma have become more apparent in the healthcare space, so has the proliferation of trauma-informed care [62, 79, 80]. The most common approach to trauma-informed care (including in HCI research [2, 23, 31]) is the Trauma-Informed Approach (TIA; [48]), a framework that originated in the Health and Human Services sphere for interacting with individuals who have experienced trauma. TIA was created by the SAMHSA [48], which emphasizes that understanding trauma on its own is "not sufficient to optimize outcomes for trauma survivors" (p.9) and advocates for approaches that prevent retraumatization. The framework is meant to be generalizable to different contexts and circumstances. TIA refrains from prescribing direct rules, and instead articulates four assumptions, six principles, and sample questions to guidance evaluation of trauma-informed services.

TIA Assumptions: The four assumptions of TIA speak to baseline qualifications of the team–or "program, organization, or system" [48]–claiming to be trauma-informed. Specifically, the team must be able to: (1) *Realize* the widespread impact of trauma, (2) *Recognize* the signs and symptoms of trauma, (3) *Respond* by applying principles of TIA, and (4) *Resist retraumatization.* SAMHSA likewise recommends "training" [48] for individuals involved in creation or administration of trauma-informed services to be capable of satisfying these assumptions.

TIA Principles: The six principles to guide application of TIA in interactions with (or "services" for) those who may have experienced trauma [23] are: (1) *Safety*: Physical and psychological safety should be guaranteed for service recipients and those who provide services. (2) *Trustworthiness and Transparency*: Trust should be built and maintained between those who receive and provide trauma-informed care; SAMHSA suggests providing "full and accurate information" and maintaining transparency [48]. (3) *Peer Support*: the development of mutual healing relationships in ways that are non-judgemental, empathetic, and voluntary. (4) *Collaboration and Mutuality*: Importance should be placed on "leveling power differences" between those administering and receiving care. (5) *Empowerment, Voice, and Choice*: recognizing each individual person's experience, and encouraging those receiving care in making choices for themselves. (6) *Cultural, Historical, and Gender Issues*:

avoiding stereotypes of trauma associated with identity characteristics while acknowledging the intersection of history and culture with trauma on a broad scale.

TIA Evaluation: Advice from SAMHSA [48] advocates for inclusion of both trauma experts and recipients of trauma-informed care in evaluation of TIA implementations. However there are no consensus "best practices" for evaluating applications of the trauma-informed approach, perhaps in part because extant approaches to evaluation exhibit inconsistency in precisely *what* is assessed regarding TIA adherence. Regarding expert or practitioner involvement, prior TIA evaluations have focused on assessing an organization's adherence to TIA principles (e.g., [10, 22, 42, 53, 104]) and staff members' knowledge of TIA (for a review, see [22]); in effect assessing trauma proficiency of a team or individual responsible for administering trauma-informed care. Regarding involvement of care recipients, evaluation has centered on a trauma-informed service's perceived adherence to TIA principles from the perspective of patients or recipients of care (without requiring expertise in TIA) (e.g., [42]). Notably, our team is not aware of evaluation techniques that measure retraumatization of those who receive trauma-informed services. Even in recent literature [24] that examines *"outcomes"* of trauma-informed care frameworks in the context of healthcare, retraumatization is not considered an outcome nor is it measured. This absence could be due to ethical implications of controlled, comparative assessment of trauma-informed artifacts ([48], p. 21).

2.1.2 Trauma-Informed Computing. While literature on TIA implies its application in clinical services "for" people who have experienced trauma, contemporary HCI literature has considered how human-centered research and design methods can be inspired or influenced by TIA. This work falls into three general categories.

Translating TIA principles to computing: Some of this research focuses on translating the six principles of TIA to HCI and specific application areas. This is most directly showcased by Chen et al.'s work on Trauma-Informed Computing [23], which seeks to adapt the principles of TIA to the computing space in general. Building on Chen et al.'s [23] Trauma-Informed Computing (TIC) principles, Scott and colleagues [95] proposed the idea of trauma-informed social media; they reflect on how the six TIA principles might be applied to improve the design and content moderation of social media, as well as company culture.

Informing research methods: Other scholars have considered how TIA can inform research methods in HCI, such as interviewing [113] and analysis of trauma-related data [106]. Additionally, Eggleston et al. [32] proposed trauma-informed usability heuristics when creating a university web portal. Some literature recommends minimizing retraumatization in research participants by opting not to directly interact with primary stakeholders/users during technology design and to explore alternative stakeholders and data in UX research–a recommendation supported by other trauma-informed design literature [1, 35, 113, 114, 118] through the use of proxies, or secondary stakeholders (i.e., experts) with ample experience interacting with anticipated users. This is exemplified by research into post-trauma regulation apps, where researchers partnered with health practitioners to discuss the implementation of TIA principles into tech design in a sustainable fashion. Similarly, Ahmadpour et al [1] designed for Social Emotional Learning of children with trauma through workshops with therapists and social workers as proxies for users, creating a trauma-informed framework for future participatory design with proxies.

Informing technology development: Prior work has also introduced technologies and prototypes claimed to be trauma-informed, although there remains scarce criteria to formally qualify a technology as adequately trauma-informed [118]. For instance, Kelly et al. [52] proposed a list of trauma-informed design heuristics to evaluate usability, while developing a portal for students who seek informational resources. Rajan and colleagues [81] develop a sexual assault reporting program; the platform itself does not directly apply TIA, but they conclude that incorporation of TIA would be essential to create a secure, accessible user experience. Saxena and Guhda [91] describe their algorithm for child-welfare cases as crafted from a trauma-informed perspective with intent to better understand parents and children, and to empower those who work cases by involving them in the co-design process.

2.1.3 Limited Adherence to TIA Assumptions and Evaluation Requirements in HCI. Formative research bridging HCI and TIA could be described as loosely inspired by TIA, rather than strictly adhering to the assumptions of team expertise, principles of TIA implementation, and expectations of evaluation by experts and recipients of care as outlined in TIA literature [23, 117, 118]. Prior HCI work does acknowledge the risk of TIA being superficially applied as a label to research and design endeavors [118], particularly when the design team does not have trauma experts or when a technology claimed as trauma-informed has not been evaluated in its adherence to trauma-informed principles.

To the former, prior efforts to apply TIA to HCI seldom clarify if team members involved in the research or design efforts have TIA training or expertise (with notable exceptions such as [83, 118].) It is therefore largely unknown if prior studies and technologies claimed as trauma-informed actually satisfy the four base assumptions of expertise.

To the latter, while evaluation of TIA implementations is explicitly recommended by SAMHSA [48], HCI literature is largely devoid of evaluation of claimed-as-trauma-informed technology in its (capacity for) adherence to TIA principles. Rather, the HCI literature intentionally or implicitly adheres to a "holistic trauma-informed approach" [83] (p. 5) in which the respective technology or artifact itself is considered a manifestation of TIA, and is thus evaluated not its in adherence to specific TIA principles but rather through context-specific criteria. An example is Ramjit and colleagues' [83] creation of a trauma-informed referral process to better coordinate survivors of intimate partner violence (IPV) with applicable services. Referral processes are described as trauma-informed through criteria specific to referrals rather than generalized trauma-informed principles, notably: referrals are trauma-informed when they are "attentive to the emotional impact of answering questions about trauma and only ask need-to-know-information required to determine fit between a client's need and the services offered" (p. 4). Likewise, evaluation of the updated referral process entailed analyzing 97 instances of the new referral form being used in its capacity to improve coordination between clinical partners, but

without acute assessment of the form's adherence to specific TIA principles or assumptions.

2.2 Data Donation and Opportunity for Trauma-Informed Design

Data donation comprises the deliberate transfer of already-created data from a person who "has" it to an entity who "needs" it [40]. It has been touted as a more consensual form of data collection relative to Terms of Service agreements and notice-and-consent prompts on websites and apps [41, 114] because data donors make conscious and active decisions to provide their data, typically by downloading it from one source and uploading it to a separate data donation app or platform (e.g., [36, 55, 77, 86]). As we unpack below, the types of data intended for donation in prior work are commonly associated with trauma, and prior work has developed various data donation process designs that prioritize donor well-being; however, an explicitly trauma-informed data donation process has yet to be articulated.

2.2.1 Donation of Data Potentially Associated with Trauma. Prior data donation efforts have targeted myriad types of data (which are not necessarily mutually exclusive), including personal data [13, 64, 96, 102], multi-person data [114], sensitive data [26, 118], and intimate data [39]. Personal data is associated with identifiable information, such as health records [59, 102]. Multi-person data represents more than one person, such as audio records collected from voice assistants where more than one person is speaking [37]. Sensitive data is a subtype of personal data that includes characteristics such as race, political stance, religious beliefs, sexual orientation, or data related to a person's sex life, among other sensitive topics. Data is deemed intimate [93] in data donation literature when it is private from others, such as data collected in personal spaces [38, 76], or related to bodily functions [37, 46].

While the word "trauma" is seldom used to describe data targeted in prior data donation literature, some examples of data donated in prior work are certainly associated with traumatic experience. For instance, past work on the donation of Instagram private message interactions sought to amass a dataset of harassment and other risks that youths face in online communication [86]. Other work has supported donation of menstrual tracking logs, which can be associated with trauma from forced sexual experiences [41] or reproductive complications, especially given fluctuating legal regulation around abortion access [4, 28, 45].

2.2.2 Data Donation Journey Design and Opportunity to Foreground Trauma. Donation of data associated with trauma posits a unique risk of retraumatization, in which a person's experiences of trauma resurface simply through the act of reengaging with data representative of that experience for donation, causing negative mental health effects [23, 106, 115, 117, 118]. Mitigation of donated-instigated retraumatization would necessitate carefully designed data donation "journeys" [23] that consider where/how retraumatization may occur and how to avoid such effects. While prior work has not proposed a holistic "trauma-informed" data donation process explicitly, the literature does contain data donation processes that are broadly supportive of donor well-being (e.g., Sensitive Data Donation [44]), as well as isolated platform design elements that are protective of donors.

The foundational elements of any data donation process are 1) the donor uploading the requisite data to the data donation platform with informed consent [37, 43, 51, 72, 73, 77, 78, 101, 111, 116] and 2) labeling or "contextualizing" the data to provide explanatory information about the data [38, 40, 41]. The actual uploading of data has often involved the downloading and uploading of consolidated data files from the source application [11, 18, 37, 38, 74, 86, 111] (thanks to data portability requirements in regulation such as the GDPR [108], or "scraping" data from the requisite account [19, 63]. Labeling/contextualizing data can vary in depth; examples include labeling donated Instagram messages as harmful or not [26] as well as semi-structured interviews with donors about their data to verbal elaboration on the personal situations and factors underpinning the donated data like menstrual logs [38].

Beyond the foundational aspects of data uploading and contextualization, data donation platforms have commonly included elements geared towards donor well-being and consent. Examples include resources for mental health and abuse [86, 117, 118], allowing donors to skip [118], and ensuring deidentification of data post-donation [86, 118]. Ensuring informed consent of donors has been of particular interest [37, 43, 51, 72, 73, 77, 78, 101, 111, 116], through explanations of a donors' data before they officially confirm their donation [40, 41] as well as visual representations of uploaded data to ensure donor understanding of its contents [37, 40]. Walquist et al. [114] proposed interfaces for procuring consent from multiple people represented in donated data. Researchers have also proposed more holistic data donation processes tailored to particular types of data being donated. Participatory data donation involves donors in co-creating research goals and working with researchers to collectively interpret data [40]. Sensitive data donation [41], rooted in data feminism [27], is a cyclical process through which donors "draw clear boundaries around" disclosure of data and serve as the contributor, collaborator, and co-creator throughout the data donation process. Our work follows in this tradition by producing a context-specific and modifiable data donation process for donation of online sexual experience data that explicitly foregrounds the recognition and mitigation of retraumatization.

3 Methodology

We produced the trauma-informed data donation (TIDD) process through a 2-year research through design effort to formally adapt the Trauma-Informed Approach (TIA) from the Public Health domain to data donation platform design. TIDD was both created and exemplified through the design, development, and assessment of Ube: a custom-made data donation app for online daters to donate data about online and face-to-face interactions with sexual partners. The intent of data donation through Ube is to amass a dataset of online content and interactions from dating apps that precede consensual and nonconsensual sexual activity (i.e., sexual violence), to ultimately yield a better understanding of antecedent patterns in computer-mediated communication that are predictive of physical sexual violence. Data collected through Ube has a high chance of being associated with sexual trauma because approximately 10% of in-person sexual violence incidents overall have

been linked to dating apps [109], in addition to qualitative evidence [6, 21, 70, 120–122]. TIDD was informed by three distinct phases:

(1) **Design and development of Ube with experts on trauma and sexual violence:** This phase entailed conscious integration of TIA principles and assumptions into data donation platform design through recurrent collaboration between an 11-member team of UX designers and researchers, software developers, and experts on trauma and sexual violence.
(2) **Observational interviews of data donors using Ube:** TIA explicitly calls for involvement of anticipated recipients of trauma-informed services in formative evaluation and improvement [48] (in our case, data donors). Accordingly, this phase involved use, assessment, and open reflection of Ube by 18 data donors.,
(3) **Reflexive thematic analysis to produce a context-specific** trauma-informed data donation process: Once Ube's design was finalized, we performed reflexive thematic analysis [16, 17] on the myriad data, recordings, and documentation produced through the prior two phases to articulate a formative TIDD process that could be adapted or altered for other data donation contexts.

3.1 Positionality Statement

Our team involved experts in UX design and research, software development, and sexual violence and clinical administration of trauma services. **User experience design and research:** UX1 constructed the interdisciplinary team and led the design of Ube and associated UX research activities. UX1 directs their university's Human-Centered Design MS program; their research agenda directly explores computer-mediated consent and unintentional sexual violence. Iterative prototyping of Ube was supported by two graduate students, UX2-3; both with academic backgrounds in Psychology specializing in sexual aggression. Interviewing of data donors was supported by UX4, a graduate student with a publication record in HCI and AI-driven harm mitigation.

Sexual violence and trauma services: NURSE1 teaches trauma-informed care at a university level, and is a director of their university's Forensic Nursing program; they are also a certified sexual assault nurse examiner (SANE) with 13 years of clinical practice. They have extensive practical knowledge of TIA and its effective application and served as the TIA "champion" [48] for this project. We also recruited SHELTER1 and SHELTER2, who are domestic and sexual violence shelter workers with seven and ten years of experience respectively. In addition to clinical expertise, we involved experts in conducting research with individuals who have experienced and perpetrated sexual violence, PSYCH1 and PSYCH2, who collectively have over two decades of experience examining risk factors of perpetrating sexual violence and women's fear of rape. PSYCH1-2 are Associate Professors in Psychology, and PSYCH2 is a co-founder of a violence and abuse resource consortium for their local community.

Software development: SD1 advised technological development of Ube; at their university they are a Founding Director of the AI Research Initiative. SD2 is a graduate student and software developer who works in trustworthy AI and LLMs, particularly in health contexts, and was the primary software developer.

3.2 Development of Ube with Experts on Trauma and Sexual Violence

Design and development of Ube began with collaboration between UX1-3, NURSE1, and PSYCH1-2 in the form of weekly meetings and milestones. During months 1-2 the team focused on identifying opportunities for retraumatization in data donation methodology and clarifying the scope of our own data donation goals. These tasks were supported by sharing professional experiences interacting with or studying those who perpetrated or were victimized by sexual violence. During the sharing of best practices, NURSE1 introduced the team to TIA, including relevant literature to review and a presentation outlining how it had been applied in other domains. We subsequently used TIA as a basis to critique and guide the design of Ube.

During months 3-4 discussion centered around the risk of retraumatization to data donors (how the act of using Ube could negatively affect their mental health through insensitive engagements with their data and forced reliving of sexual trauma from online dating). Initial designs of Ube took the form of text-based scenarios to put a spotlight on the precise wording of data donation prompts and other inquiries of the donor to both identify and rectify potential retraumatization triggers.

During months 5-6, SD1 and SD2 joined weekly meetings to ensure technical feasibility and confirm specific design requirements. The output of these months included intricate key path scenarios: interface mockups depicting every possible screen that a donor could see in their interactions with Ube, and text explaining how and why one would arrive at that screen. Opportunities for mitigation of retraumatization likewise became more obvious with increased fidelity and specificity in Ube's data donation journey.

In months 7-8, two sexual and domestic violence shelter workers (SHELTER1-2) joined the team for six cognitive walkthroughs [99] of our Ube prototype. These were moderated by UX2 and UX3, and each took three hours. Rather than merely inspecting for usability challenges, goals of these cognitive walkthroughs were to identify latent risks of retraumatization and opportunities for better adherence to trauma-informed principles. One final (seventh) walkthrough was conducted with SD1-2 to validate the technical feasibility of Ube.

Months 7-12 centered on the development of Ube as a functional mobile app for Android devices by SD1-2. All members of the team, aside from SHELTER1-2, assessed development and ensured the design aligned with the Figma prototype. Concurrently with the technical development of Ube as a functional app, and progressing through month 15, the team engaged in retrospective reflection into how and why we qualified Ube's design as trauma-informed, including the use of reflexive thematic analysis (RTA) for conscious explication of how TIA's assumptions and principles were exemplified in Ube.

3.3 Observational Interviews with Data Donors While Using Ube

Proper application of TIA necessitates "evaluation" of services or artifacts [48], particularly to "solicit feedback from people who use [the trauma-informed] services" [48]. This was reinforced by our team member NURSE1, an expert in trauma-informed care. We

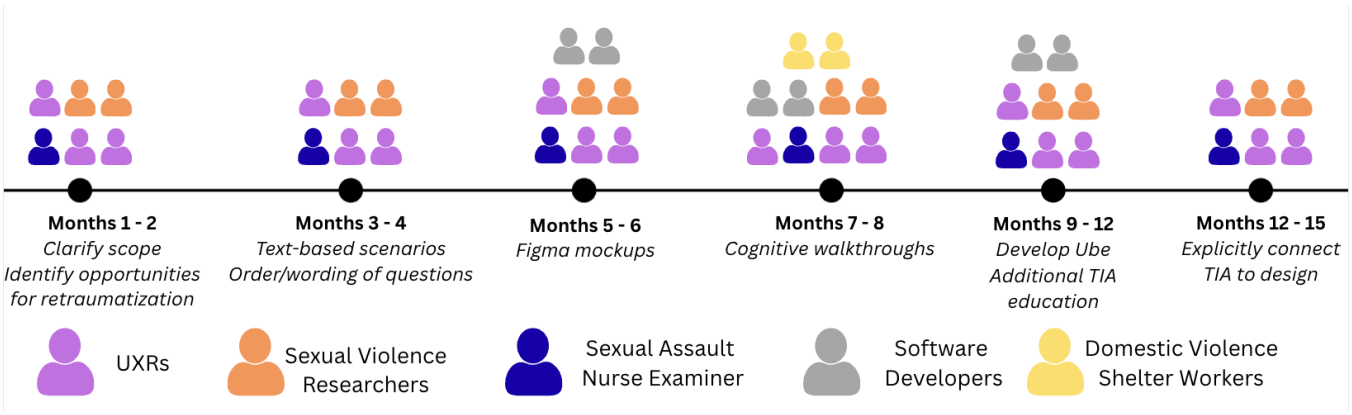


Figure 1: ***A map of collaborator involvement in Ube design and development over the first 15 months.***

subjected Ube to assessment with 18 data donors through interviews and observations while they used Ube to make data donation choices about their online dating sexual experiences. Interview conduct, subsequent modification to Ube, and thematic analysis to articulate TIDD was iteratively performed through months 16-24.

The HCI literature acknowledges a lack of evaluation of trauma-informed technologies and artifacts by survivors of the focal trauma (e.g., [1, 83]). There is thus limited prior work within HCI that our team could use as the basis for our own trauma-informed evaluation of Ube. There are some existing trauma-informed evaluation approaches from the health and human services sphere, however they are not readily applicable to our assessment of Ube, or to trauma-informed technologies more generally. The biggest challenge to adapting extant evaluation approaches for TIA adherence is their intention to evaluate *services* for trauma-informed care, which typically involve a health professional interacting with a client in a clinical setting to provide care for experiences that may have incurred trauma. Associated evaluation methods assess staff and organizational knowledge of trauma [10, 22, 53, 104], and clients' perspectives on how well the care they received adheres to trauma-informed principles [42]. These approaches mainly take the form of quantitative scales/surveys, with questions making reference to *"staff"* and other terms that fit a clinical context, but less so technologies designed through a trauma-informed lens.

Another challenge with adapting these prior evaluation approaches for Ube is that users of our data donation platform would not be using our application *for* care, thus making inapplicable any evaluation approach centered on the quality of care received. Relatedly, Ube was not designed for exclusive use by sexual violence survivors, and it deliberately avoids any language that may imply one is a survivor. Incorporating evaluation approaches that explicitly ask about Ube's alignment with trauma-informed principles would be incongruent with Ube's efforts to avoid labeling of users' experiences because explicit questioning about trauma and TIA adherence would risk implying that Ube users did have, or should have had, an experience potentially triggering of trauma.

In considering alternative approaches to evaluating Ube's adherence to TIA principles, we noted how TIA literature encourages use of questions and language during evaluation that are "familiar" and "appropriate" to the "individuals being served" [48], which may differ starkly from how professionals well versed in TIA may communicate. We thus opted for an interview protocol that involved donors freely engaging with the data donation interface and thinking aloud about their donation decisions and general reactions to all interface prompts and questions. Assessment of Ube by data donors, particularly their positive and negative feedback on specific design choices, served as support or criticism of our attempts to manifest underlying TIA principles and assumptions. For example, donor criticism of an interface feature that was intended to adhere to the TIA principle of cultural, historical, and gender issues would be an indication that we had failed to effectively translate that principle in design.

We routinely asked about comfort levels while progressing through the data donation journey and verbally reiterated that donors could discontinue the data donation process at any time or skip any data donation prompts and questions in the interface without adverse consequence to their financial compensation. Observational interviews were conducted synchronously online via Zoom, ranging from 94 - 210 minutes. All sessions were audio- and screen-recorded.

3.3.1 Interview Participants and Recruitment. Recruitment took place through Craigslist, University mailing lists, and public social media posts. Participants were compensated with a $30 gift card. The recruitment message explained that participants would use a data donation platform that collected data about their online dating experiences to help researchers better understand the connection between dating app-use and sexual activity, including sexual violence. Participants were informed that they would be asked (not required) to provide data involving their messaging conversations and sexual encounters. Our University's Institutional Review Board (IRB) approved the study.

Participants were between the ages of 19 and 44 and were from the United States. Half (9) of the participants reported experiencing sexual harm or harassment (i.e., trauma) either online or in person from a person met on a dating app, though self-identification with sexual harm was not mandatory due to the need for data about online interactions that precede both consensual and non-consensual sex to train risk detection AI [25, 86]. See Table 2 in Appendix A for full demographic reporting.

3.4 Data Collection and Analysis to Produce Trauma-Informed Data Donation

We used reflexive thematic analysis (RTA) [16, 17] for articulating a formative TIDD process from design, development, and assessment of Ube. We chose RTA because of 1) its flexibility in data sources that can be subjected to analysis [16, 17], and 2) flexibility in theoretical frameworks that can be consciously applied during analysis [16, 17] (in this case, TIA). Data incorporated into RTA came from the two prior phases. First was the expert-informed design and development of Ube, during which TIA principles and assumptions were consciously mapped to specific design choices in Ube with trauma and sexual violence experts. The second was use of Ube by data donors, through which their reflections on specific interface design choices served as support or criticism of our attempts to manifest underlying TIA principles and assumptions.

Data subjected to analysis from the donor interviews included transcripts. Data from the expert-informed design and development of Ube was more diverse, including text-based key path scenarios, Figma prototypes, weekly meeting and design notes, and cognitive walkthrough recordings and transcripts. All data underwent reflexive thematic analysis (RTA) [16, 17] to produce the TIDD process, led by the first author and involving UX1-3, PSYCH1-2, and NURSE1. RTA involves six steps [16, 17]: 1) familiarization with the data; 2) initial coding; 3) generating initial themes; 4) developing and reviewing themes; 5) refining, defining and naming themes; and 6) writing up results. We used a combination of Miro boards, collaborative text documents, and spreadsheets for shared coding and thematic generation.

4 Introducing Ube: A Data Donation Platform for Online Dating Sexual Experience

4.1 Long-Term Project Details

Ube is a data collection instrument that is part of a long-term project that ultimately aims to 1) amass large-scale empirical evidence of how dating app-use shapes (mis)understandings of consent to subsequent physical sex acts, and 2) generate computer-mediated and data-driven solutions to sexual violence that correct misperceptions of consent across online-to-offline sexual interactions that can supplement or replace safety technologies for in-person meetings.

Ube specifically focuses on the potentially-harmful role of computer mediation in sexual consent practices; in this case, how the process of using a dating app to discover and interact with a sexual partner online can influence how individuals provide and (perceive to) receive agreement to sex in subsequent physical settings. This is motivated by burgeoning evidence [65, 82, 97, 119, 121] that dating apps and associated technologies can (mis-)shape the ways that consent to sex is understood, potentially leading users to perpetrate sexual harm without conscious intent. Examples in past work include assuming consent to sex through indirect cues received online such as physically revealing photos and emojis, [97, 119, 121] as well as misunderstanding one's own ability to decline sex by virtue of having met on a dating app commonly associated with sexual intent [54, 65, 97, 121].

Most immediately, Ube's data will be used to create novel forms of sexual risk-detection AI. Previous sexual risk detection AI efforts have been trained on publicly available datasets (for a review, see [87]), a practice scrutinized due to concern that sexual abuse online often takes place in private channels [87]. Yet private datasets which represent personal messaging exchanges, for example, are difficult to construct due to concerns around incidental retraumatization of victims [87], and difficulty obtaining enough data. Ube fills these gaps by creating a dataset of personal messaging exchanges and related in-person sexual interactions, both consensual and nonconsensual.

Aside from data availability, the premise and implementation of risk detection AI has received critique for susceptibility to biased profiling [92] and subsequently inequitable outcomes. Yet risk detection AI fueled by Ube data will not follow traditional punitive responses to sexual risk detection such as removing users from the platform. There are many issues surrounding these tactics, such as possibility that a person is mislabeled as a perpetrator and improperly punished, or the possibility of harmful behavior remaining unchanged after a punitive response. Rather, Ube data-driven sexual risk detection AI will pursue tailored, corrective solutions. These are planned to include personalized consent educational interventions when patterns of problematic consent practices are detected, and messaging interaction scaffolding that nudges online daters towards, for instance, transparent dialogue about expectations so as to preemptively correct misunderstandings of expectations for sex.

4.2 Ube's Technical Details

Ube is a standalone app for Android devices, developed using React Native and JavaScript, with dependencies managed through NPM. Ube does not integrate with or depend on a specific dating app. (Interface design examples of Ube are presented in conjunction with TIDD in the next section.)

Ube works toward project goals by collecting digital trace data [14, 15, 40] in the form of screenshots from online daters that represent how they conveyed their openness to sex online prior to meeting, or how they perceived their partner to have conveyed openness to sex online. These screenshots represent messaging interactions that precede in-person sexual meetings, including from dating apps, text messaging, and other social platforms. Donors also perform data contextualization through open-ended text responses to 1) explain how their donated messaging interaction screenshots represent how they conveyed or perceived openness to sex online and 2) how consent was (not) exchanged during subsequent attempted or completed physical sexual activity.

5 Findings

Our analysis produced a Trauma-Informed Data Donation (TIDD) methodology, exemplified by Ube, a data donation app for donation of online-to-offline sexual experience data from online daters. TIDD is a process of six phases that simultaneously support donation of data indicative of sexual experience with online daters, while also seeking to mitigate retraumatization that the act of data donation could instigate. Each of the six phases of TIDD is informed by specific trauma-informed assumptions and trauma-informed principles from the Trauma-Informed Approach (TIA; see section 2.1.2 for a review of the full list of principles and assumptions underpinning

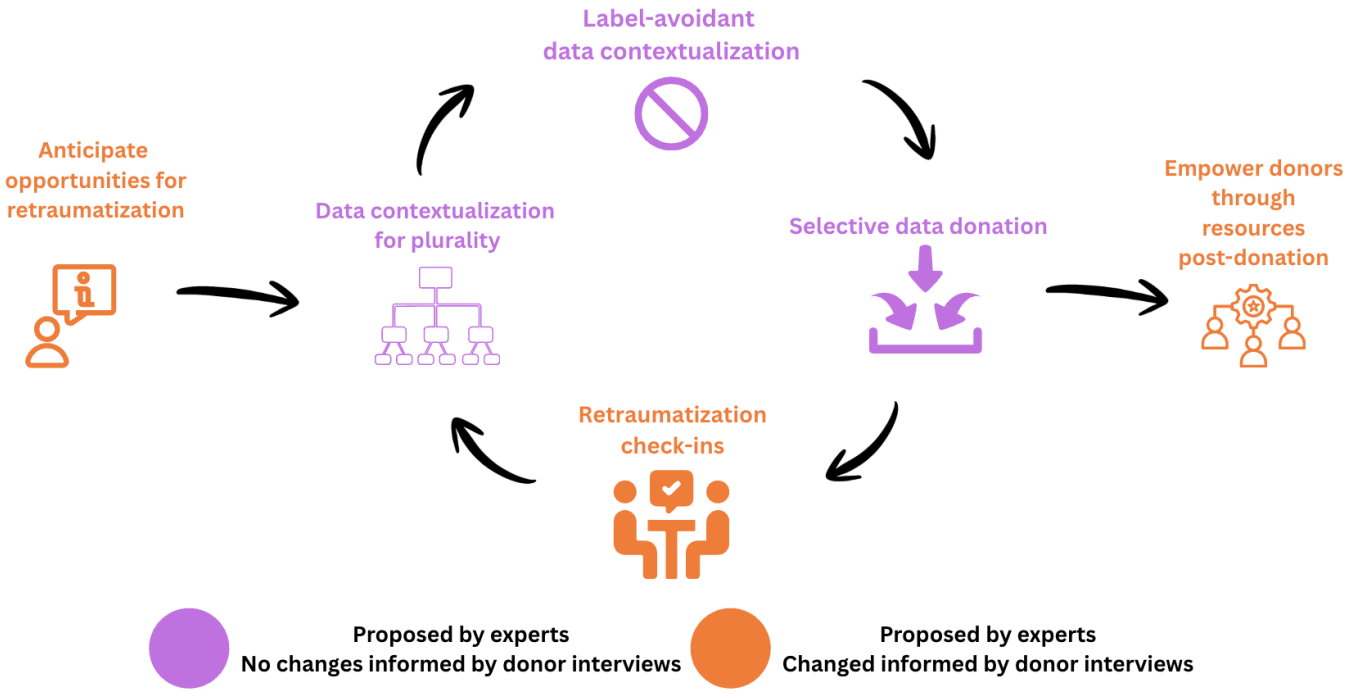


Figure 2: ***The trauma-informed data donation process. The middle four phases are cyclical and can be repeated multiple times in a donor's data donation journey.***

TIA). In section 5.1 we give a high level walkthrough of our data donation app Ube to demonstrate how the app's design exemplifies each phase of TIDD. Then sections 5.2-5.7 each focus on a specific phase of TIDD.

5.1 Exemplifying Trauma-Informed Data Donation with Ube

Figure 3 provides a high-level walkthrough of a data donor's progression through our data donation app called Ube. Each interface in the walkthrough exemplifies an underlying phase of TIDD. The figure does not depict ***Phase 1****: Equip donors to foresee retraumatization*, as it takes place outside of the data donation app, prior to donation. We operationalized that phase through recruitment materials and communications with potential donors prior to accessing Ube. For instance, Phase 1 was created with Principle #5: **Empowerment, voice, and choice** in mind; recruitment materials specified that participants would be answering detailed questions about their sexual experiences in online dating in order to empower donors to make informed decisions about their own risk and willingness to participate.

Interface (1) depicts ***Phase 2****: Data contextualization for plurality.* Donors interact with Ube through a chatbot-style interface by first answering sequential questions about the sexual experience to be donated about, each of which enables increasingly-personalized data donation and contextualization requests as the donation journey progresses. The chatbot design was selected specifically because it allows pluralistic data donation journeys by making donation a sequential process of questions, with donor responses each affecting what they are subsequently asked and how. We mapped this to Principle #6: **Cultural, historical, and gender issues** because it allows Ube to avoid questions that are not just inapplicable to the donor, but also potentially insensitive or inappropriate for their particular sexual experience.

The top part of Interface (2) exemplifies ***Phase 3****: Label-avoidant data contextualization.* Once a specific sexual experience is identified, Ube asks contextualizing questions about how consent was given and perceived to be received. In this example, Ube asks the donor how they knew the other person was "open to sexual contact." We deliberately used this wording across questions instead of "give consent to sexual contact" to avoid learned social scripts about how consent is supposed to be exchanged and also inadvertent or implied accusations that the donor may have practiced consent exchange incorrectly, which we tethered to both Principle #1: **Safety** and Principle #5: **Empowerment, voice, and choice**.

Interface (2) also depicts ***Phase 4****: Selective data donation.* In this example, enabled by prior contextualization questions, Ube asks the donor to upload a screenshot of the messages sent or received on the dating app that helped the donor understand that the other person was "open to sexual contact." This contrasts with more passive and general forms of data donation, such as having a user download a JSON file representative of all of their messaging interactions from a dating app (per data portability functionality) and donate the entire file. This choice aligned, in part, with Principle #2: **Trustworthiness and transparency** because donors would

Table 1: The phases of Trauma-Informed Data Donation (TIDD) are defined, exemplified in Ube's app design, and mapped to assumptions and principles from the Trauma-Informed Approach.

Phase (see Figure 2)	**Definition** (see sections 5.2-5.7)	**Operationalized in Ube** (see Figure 3)	**Trauma-Informed Assumptions** (see section 2.1.2)	**Trauma-Informed Principles** (see section 2.1.2)
1. Equip donors to foresee retraumatization	Inform donor of details surrounding data donation so they can anticipate trauma re-exposure	Prior to opening the Ube app, donors are informed of use and recipient of donated data, type of data they will be asked to donate, and data protection measures	**Resist** retraumatization by supporting donors in anticipating such risk	**2:** Trustworthiness and Transparency **5:** Empowerment, voice, and choice
2. Data contextualization for plurality	Trauma cannot be recognized and responded to appropriately with a "one-size-fits-all" data donation journey	Chatbot-style inquiries about donor's sexual experience are used to personalize their data donation journey and avoid prompts that could be insensitive or irrelevant	Preemptively **Recognize** and **Respond** to potentially traumatic experiences being reported	**5:** Empowerment, voice, and choice **6:** Cultural, historical, and gender issues
3. Selective data donation	Support donation of selectively chosen data rather than passive "data dumps" and ability to change donation decisions	Donors curate their data off platform and donate select portions of online dating messages; donation deletion is allowed	**Resist** retraumatization through user autonomy over the data they donate	**2:** Trustworthiness and Transparency **5:** Empowerment, voice, and choice
4. Label-avoidant data contextualization	The data donation platform should not judge or define a donor's experience, even implicitly	Behaviorally specific wording avoids labeling donor as victim or perpetrator of sexual harm	**Resist** retraumatization by avoiding labeling of donated experiences	**1:** Safety **5:** Empowerment, voice, and choice
5. Retraumatization check-ins	Incorporate assessment of donor's well-being during the data donation process, and confirm their willingness to continue	Ube incorporates recurrent "check-ins" to identify retraumatization incurred by data donation	Actively **Realize** the traumatic impact of data donation	**1:** Safety **4:** Collaboration & mutuality
6. Retraumatization management after donation	Conclude data donation with trauma-related resources	All donors are provided a resource list for sexual trauma services after donation concludes	Actively **Realize** and **Resist** the risk of retraumatization post-donation	**1:** Safety

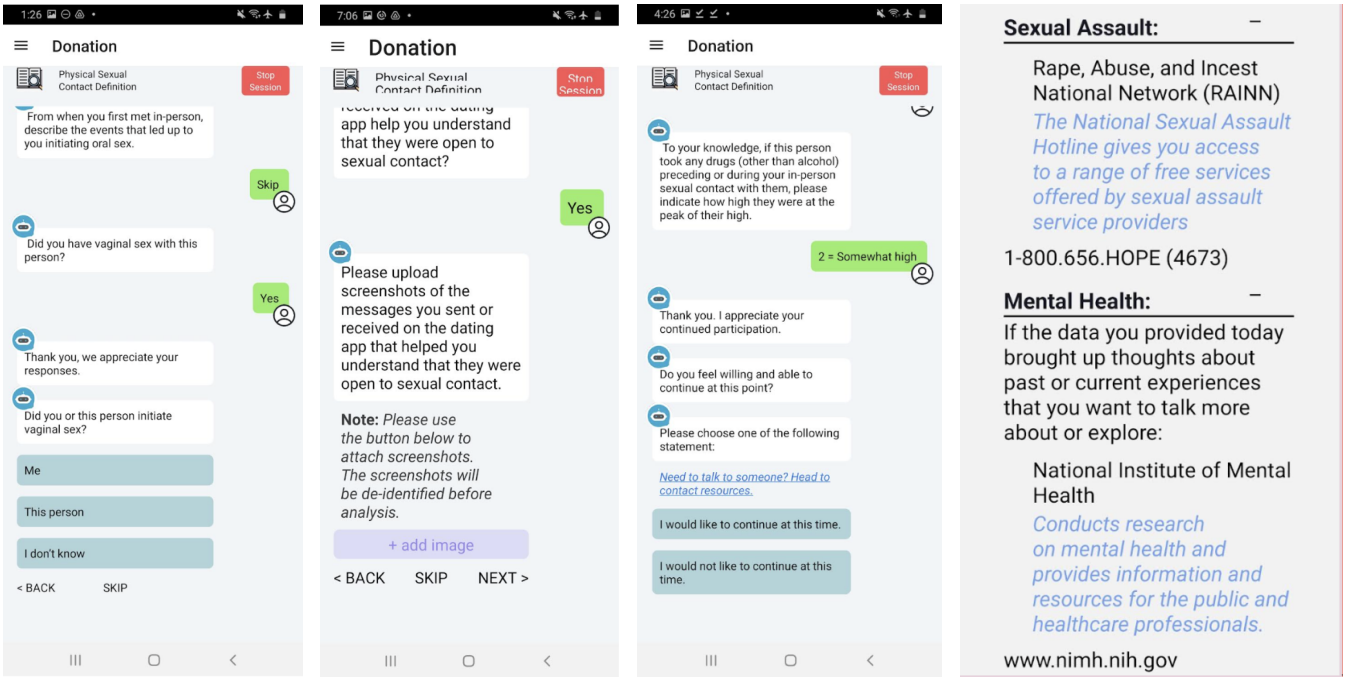
be fully aware of what online dating data Ube did and did not have from them.

Interface (3) depicts ***Phase 5****: Retraumatization check-ins.* Prior to questions or prompts that our trauma experts identified as particularly sensitive, Ube asks, "Do you feel willing and able to continue at this point?" Those who select that they would not like to continue are provided with a resource list curated by sexual violence and trauma experts (also depicted in interface 4) and exit the data donation process. This check-in was designed with Principle #1: **Safety** in mind to enable identification of retraumatization mid-donation and subsequent discontinuation of the donation process, and Principle #4: **Collaboration and mutuality**, by providing donors with a choice over whether they continue participation rather than defaulting to ending donation.

Interface (4) represents ***Phase 6****: Retraumatization management after donation.* At the conclusion of every donation, regardless of responses, donors are provided with a resource list curated by sexual violence and trauma experts on our team. This was one of the first interfaces designed with TIA in mind, and was mapped to Principle #1: **Safety** from its inception.

5.2 TIDD Phase 1: Equip Donors to Foresee Retraumatization

TIDD enacts the trauma-informed assumption of Resisting retraumatization by equipping data donors with information to help them personally anticipate (and potentially avoid) the risk of retraumatization through data donation *before* opening and interacting with the data donation platform. While trauma-applicable information will surely vary in each donation context, we identified the following information as particularly relevant in our case of sexual experience data donation through interviews with donors: the type of data to be provided, the purpose of data collection, the recipients of the data, data protection measures, and resources that will be provided to donors during and after donation. This can be augmented with the affordance to interact with the researchers or moderators of the data donation platform, such as through email, private messaging, or voice calls, to broach and answer additional questions. This transparency can build trust with donors (enacting principle #2 of TIA: Trustworthiness and transparency) and empower donors to make informed choices to begin data donation (enacting principle #5: Empowerment, voice, and choice).



(1) Data contextualization for plurality (2) Label-avoidant DC & Selective data donation (3) Retraumatization check-ins (4) Retraumatization management

Figure 3: A high-level walkthrough of Ube's data donation process mapping to phases of Trauma-Informed Data Donation.

5.2.1 Initial Ideation with Trauma and Sexual Violence Experts. The concept of preparing donors for their participation first arose during cognitive walkthroughs with our trauma and sexual violence experts. They expressed concern that the sensitivity of the topics being discussed in Ube could be alarming to those participating if they were not warned beforehand, and at worst retraumatizing due to their unexpected occurrence.

SHELTER1: *"[The contents of data that will be collected] would just be something that I would also recommend being upfront about of like, hey, this could bring up maybe things that past experiences that you didn't identify, or that might bring up painful past experiences. Like if that happens, we have these resources available in the app."*

Most of the discussion surrounding this phase of TIDD pertained to preemptive information about trauma and mental health resources, under the premise that knowing such resources would be available during donation may better inform initial decisions to start donation. Per NURSE1: *"Obviously, one of the risks of the study in and of itself is that you might be triggered or traumatized. So therefore, we have these [resources] available to you. So I do think it's important to have that early on. And then at some point, I, I, I would like us to be more proactive, then expecting the participant to try to remember and figure out that they need to go to the hamburger icon, I think we need to make these resources and supports more readily available [when they first access the platform]."*

At the conclusion of the expert-involved design phase, Ube included an introduction screen within the app that provided ready access to mental health resources whenever the donor wanted. Prior to accessing the data donation platform we also had a consent form that included information on the type of data to be provided, the general basis of questions and donation prompts in Ube, and data collection procedures.

5.2.2 Iteration Informed by Interviews With Data Donors. Donors provided ample evidence that information needs to be preemptively conveyed to allow them to foresee retraumatization. What we did not anticipate was how the absence of applicable information can, in itself, generate retraumatization triggers through speculative "gap filling." During interviews, donors fixated not on the resource-laden introduction page of Ube, but on the relatively scarce information about the precise format of data to be donated, along with immediate and potential future recipients of donated data. They openly pondered, and worried, about how their data would be used–in effect, expressing a retraumatization concern related to potential loss of control over the narrative and use of their sexual experience.

While this information was in our consent form, to a degree, donors expected this content to be reiterated prior to, or during initial access to, the data donation app. Donors commonly paused during the donation process and openly broached questions (or expressed surprise) around the nature of curating and donating data, and who precisely would receive the data. While engagement with researchers during the interviews helped manage these concerns, this would not be a practical way to mitigate retraumatization related to unknown (mis-)use of donated data during large-scale deployment of Ube.

Through these concerns we learned that equipping donors to foresee retraumatization involves more than just preemptive resource access, but extensive information about the data to be donated, who will receive it, and how it will be used. In response we built more elaborative recruitment and priming materials based on how we answered such questions during interviews (beyond baseline criteria of an IRB-required consent form) to ensure that retraumatization concerns would be based on accurate (as opposed to speculated) understanding over use of data.

5.3 TIDD Phase 2: Data Contextualization for Plurality

TIDD then Recognizes and Responds to potential traumatic experience (TIA assumptions 2 and 3) by personalizing each donor's data donation journey. Like trauma-informed care practices in typical clinical settings that necessitate personalized approaches to each patient, this TIDD phase acknowledges that there should not be a "one size fits all" approach to data donation, but rather a plurality of potential "journeys" through the data donation process. This is performed by asking contextualization questions about the experience being donated, early on in the donation journey, to avoid questions, prompts, and donation requests that do not apply, and could be potentially insensitive or triggering of retraumatization. This enacts TIA principle #6: Cultural, historical, and gender issues, particularly through avoidance of assumptions and questions that have historically undermined the sexual violence accounts of victims [67, 84, 89]. Conversely, personalization of data donation journey also provides donors with increasingly granular opportunities to contextualize their data in their own words without the cognitive fatigue of unrelated prompts and data demands, employing TIA principle #5: Empowerment, voice, and choice.

5.3.1 Initial Ideation with Trauma and Sexual Violence Experts. This phase emerged through discussion of the different types of sexual experiences that would necessitate very different lines of inquiry in Ube based on the professional experiences of our expert colleagues when interacting with victims and perpetrators of sexual violence. Early meetings involving UX1-3, PSYCH1-2, and NURSE1 helped the team identify 12 different types of sexual experiences that may be donated, each necessitating different questions, prompts, and data to be donated. These experiences varied based on factors such as who initiated the sexual act, whether the act was wanted by the donor or not, and whether the act was perceived to be wanted by the other person, among others. The team discussed the necessity for Ube to respond differently to donors based on these experience types, which first necessitates identification of which type of sexual experience is being donated.

We supported pluralistic data donation journeys in Ube through a chatbot-style interface that asks successive contextualization prompts, each serving to tailor subsequent questions and data donation prompts. Cognitive walkthroughs among our team members were pivotal in identifying nuanced variations in data donation journeys based on granular aspects of sexual experience. Per SHELTER2 in advocating for revised wording in a contextual prompt for donors reporting an unwanted sexual experience: *"Maybe they weren't able to let the other person know [that they didn't want to have sex], even if they wanted to. So maybe just leaving [the contextualization prompt] a little more open ended."* Similarly, in another cognitive walkthrough PSYCH1 was asked to imagine they were donating a specific sexual experience. PSYCH1 commented, *"I think it seems like that first [prompt in the interface] assumes that I had a desire to have sex with them. But in this case I did not."* These cognitive walkthroughs repeatedly reminded us how many different data donation journeys needed to be supported and, with it, the designed capacity for Ube to identify which journey a given donor should follow.

5.3.2 Iteration Informed by Interviews With Data Donors. Fortunately no donor expressed disagreement or confoundment over a data donation or contextualization prompt, suggesting that our identification of pluralistic data donation journeys and associated contextualization prompts was comprehensive during cognitive walkthroughs with our team. Yet some donors were unsure why they were receiving "a lot" of contextualization prompts, indicating that donors are not necessarily aware that these prompts help tailor the subsequent data donation journey. In some cases donors requested that the data donation journey be shorter. Per P1: *"The problem I was noticing with your chat [interface], as it currently is, is that, that a lot of the questions are the same. And like, I could see myself getting bored."*

In actuality Ube did not have any repetitive prompts, but P1's comment about contextualization prompts being "the same" taught us that the nuanced differences and purposes for each prompt are not always understood. Others suggested that the utility of contextualization prompts for personalizing the data donation journey could be made more apparent. Per P14: *"Just say, like, this is vital to our research or something."*

5.4 TIDD Phase 3: Selective Data Donation

TIDD seeks to Resist retraumatization (TIA assumption #4) by providing donors with full autonomy over what precise data they do (not) donate, and thus the ability to withhold their data if its provision to the data donation platform (and researchers operating it) induces new stress associated with the representative sexual experience. We define "selective" data donation as: 1) the curation of one's data off-platform, thus ensuring that the only data actually uploaded to the data donation platform is strictly that which the donor wants to provide, and 2) capabilities to manage donated data, including further editing and deletion. The intent of these selective capabilities is to retain donor control over their data (employing TIA principle #5: Empowerment, voice, and choice), which also intends to build a donor's trust in the data donation process (employing TIA principle #2: Trustworthiness and transparency).

5.4.1 Initial Ideation with Trauma and Sexual Violence Experts. Decisions over how to scaffold data donation through Ube's interface–paticularly decisions over supporting donation of packaged JSON or similar files exported from dating apps, or more labor intensive donation of manually curated data–occurred over the course of months 1-6 and involved UX1-3, PSYH1-2, and NURSE1. We opted against the streamlined approaches of exporting/importing files from dating apps representing all of a donor's online dating interactions, on the basis that it clashes with TIA's fundamental principles

around choice and transparency. As identified in other data donation research [37, 43, 51, 72, 73, 77, 78, 101, 111, 116], donors may be challenged with understanding the contents of large, prepackaged files of their dating app account data and they may be limited in choosing which specific parts of that data they do and do not feel comfortable donating.

Our selected approach had donors manually curate their data outside of the data donation platform by taking screenshots of messaging/online interactions preceding the sexual experience they are donating about, thus giving them full control over what messages or portions of the interaction they wish to share. While we recognize prior data donation work that involves donors visualizing and reviewing uploaded data before confirming donation [37, 40], we sought to avoid any uploading of data to Ube unless the donor had express intent to donate it. Recognizing that donors may face inherent limitations with creating screenshots that selectively include/exclude specific messages, we did incorporate a screenshot editing feature within Ube, which lets donors visually edit and obstruct aspects of their donated screenshots (e.g., blurring out specific words or portions of messages). Additionally, the landing page of Ube included a section dedicated to deletion or editing of prior donations.

In an effort to prioritize donor understanding of the data they provide, and increase granularity in contextual information about the data, Ube requested data pertaining to individual contextualization prompts (thus a donor could be asked to donate different data at several different times in the data donation journey). We found selective data donation to be tightly linked to TIDD phase 2 (pluralistic data donation journeys) in this regard because we could use contextualization prompts to ensure donors were asked for data only directly relevant to their sexual experience.

5.4.2 Iteration Informed by Interviews With Data Donors. Some donors correctly deduced our intentions for selective donation in their open reflections on screenshot-oriented data donation prompts. Per P4: *"If I was just typing or submitting information in just one file, I would miss some information, and there are some advantages and disadvantages of having some questions asked."* This slower, intentional process allowed donors to thoroughly reflect on their decisions to donate at each point, which often culminated in decisions not to donate data. While this may appear counterintuitive (a data donation process that seemingly discourages donation), but from a trauma-informed lens this is an acceptable and even laudable outcome because it foregrounds donor choice. Several donors openly reflected on their decisions not to donate data. Responding to a request for a screenshot of a conversation between themselves and their sexual partner, P5 responds: *"Well, I think this one is disturbing. Yeah, I believe the conversation was private. And I should not disclose any kind of information to a third party."* Interviews produced additional examples of donors who opted not to donate screenshots, but were otherwise open to continuing the data donation journey and answering data contextualization prompts. Per P15: *"I wouldn't be comfortable doing that [donating screenshots]. I feel like those are more private, so like they're really, really private. So for me. It doesn't really matter if it's all part of the research. But like I can explain those kind of situations [through free-text responses to contextualization questions]."*

5.5 TIDD Phase 4: Label-Avoidant Data Contextualization

While TIDD uses data contextualization prompts to personalize data donation journeys, it also seeks to Resist retraumatization (TIA assumption #4) by avoiding labels on donors' experiences in such prompts. In Ube's case, this entailed avoiding terms like "victim," "perpetrator," and "sexual violence" in prompts to donors because doing so could risk labeling, or encouraging donors to self-label, a sexual experience as wrong or harmful even if they do not subjectively identify the experience in that way. Losing agency over one's experience, or worse, being told that one has done something wrong or is actually the victim of a wrongful action, can incur its own trauma (for example, people who have experienced sexual violence sometimes do not file reports to avoid being labeled as victims [50]).

Labels can be avoided through the use of "behaviorally specific" and objective wording free of any negative and positive connotations, as we were taught by NURSE1. For instance, instead of asking a donor if they were a "victim of sexual violence," we would instead ask if they experienced a sexual act that they did not want. This protects donors who do not self-label as a victim (enacting TIA principle #1: Safety). It also keeps the perspective of the donor as a capital requirement, and ensures that those collecting the data remove their perspective as much as possible (enacting TIA principle #5: Empowerment, voice, and choice).

5.5.1 Initial Ideation with Trauma and Sexual Violence Experts. Trauma and sexual violence experts on our team were alarmed by the early use of overt harm-related language in formative Ube design concepts from UX1-3. Per NURSE1 during an early review of data contextualization prompt phrasing: *"As soon as I read the question, I don't know if you saw my face, I went, like, [gasping face] that's very jarring to move from that to that. Like what! No, I did not sexually harass this person."*

PSYCH1-2 discussed the common use of behaviorally specific wording for measuring experiences of sexual violence called the Sexual Experiences Survey (SES; [56, 57]). For example, the perpetration scale asks, *"Since the age of 14, have you ever threatened to physically harm a woman or someone close to her in order to have oral sex with her or make her have oral sex with you?"* The idea behind use of behaviorally specific wording in a perpetration context is that participants may not view their experiences as perpetration. The notion of a data donor being unexpectedly called a perpetrator of sexual violence during use of Ube was universally rejected by the team.

Assigning donors the label of victim also carried risk of retraumatization. In the words of NURSE1, *"It's not our decision to make for them, either, because this person might not be traumatized that this happened to them. It might not impact them psychologically."* Labels may also disempower and frustrate donors who feel they are losing the choice to define and contextualize their own experiences, directly clashing with TIA principle #5: Empowerment, voice, and choice. Per NURSE1 from another meeting: *"If I didn't feel victimized, and somebody was calling me a victim, I would be like what the hell, right, like [...] I get that, too."*

5.5.2 Iteration Informed by Interviews With Data Donors. Donors did not outwardly experience any discomfort with the behaviorally specific wording used in Ube's prompts and questions, which we expected given that our wording choices were based largely on phrasing that our trauma and sexual violence expert team members used in their own professional roles. Some donors did express reluctance to describe their sexual experiences in Ube generally speaking, which was an allowable outcome given the capacity to decline any data donation and contextualization prompts in *Ube*. P7 reflected on their response to a question that used behaviorally specific language: *"I guess I just don't want to describe it. [...] I don't think I want to talk about it. I guess it's a little corny or embarrassing."* Other donors openly discussed feeling comfortable throughout their donation experience. Per P3: *"I don't think there is any part that gives me a wrong impression…for me, I can say there is no fact that I feel is offending."*

We did learn of opportunities to be more efficient with our behaviorally specific wording though, which informed iterations to our phrasing within Ube. Some donors felt as though data contextualization prompts took a long time to get to the crux of the issue, as P13 reflected: *"You're beating around the bush. Say it. So if you want more, you're just gonna have to ask. [...] But for me like, I'm just gonna give you the same term you gave me, just because it seems if that's how the questions are asked, I'm gonna specifically answer it a certain way."*

5.6 TIDD Phase 5: Retraumatization Check-Ins

Despite best efforts, retraumatization can never be fully mitigated. TIDD can adhere to the trauma-informed assumption of Realizing the impact of trauma during data donation by incorporating active assessments of a donor's comfort during the donation journey and their willingness to continue as to a way to identify emergent retraumatization. Allowing donors dedicated opportunities to reflect on their willingness to continue data donation enacts TIA principle #4: Collaboration and mutuality), while mutually reinforcing emotional safety for our donors (TIA principle #1 of TIA: Safety). While this phase does not prescribe specific moments in a data donation journey where retraumatization check-ins should be interspersed, through consultation with donors, their presence in Ube was ultimately placed 1) after acknowledgments from the donor of having experienced an unwanted sexual act and 2) before passages of donation and contextualization prompts that our trauma and sexual violence experts considered particularly sensitive.

5.6.1 Initial Ideation with Trauma and Sexual Violence Experts. The retraumatization check-in was born out of a conversation around helping strategies that may replace filing a report when a donor communicates an experience that could be considered sexual assault. Per NURSE1: *"We agree like [when a donor reports they are a victim] this isn't a mandatory report, I understand that. But like ethically, as humans and me as a health care provider like to just not extend anything to them. That's where I struggle."* To cope with this dissonance, and replace the act of a formal report, another expert suggested periodic check-ins that allow a donor to make a self-determination of comfort to continue, particularly after answering data contextualization prompts in ways indicative of sexual assault.

Per SHELTER2: *"I don't know if that's possible to do like a check in with the participant, because I feel like at this point, then you'd be like, oh shit, we're talking about like, an assault."* We initially decided to intersperse these check-ins randomly so as to avoid implying to donors that their donated experience was somehow harmful or deserving of a stop in donation. The check-ins asked donors whether they felt comfortable continuing, allowing them to select that they either were or were not. Donors who selected that they were not okay with continuing received a resource list, curated by our team's trauma and sexual violence experts.

5.6.2 Iteration Informed by Data Donors. Overwhelmingly, donors appreciated the retraumatization check-ins. Per P6 after reading the check-in aloud: *"That's cool, too, that it checks in with you. because it's not just so heavily focused on collecting the data. It's also kind of paying attention to how the person is feeling about it."* This reflected a common view of the check-ins as a positive intrusion in the data donation journey, even if donors did not need or want to stop their donation. In the words of P11: *"Sick. I think it's just good to have those just in case, because I know that stuff like this could be scary [data and questions about sexual experience]. So I think that's good. I really appreciate having all of that."*

Although the check-ins were appreciated, some donors assigned meaning to the placement or timing of their occurrence. For instance, P4 received a check-in early in their data donation journey and immediately speculated on what this meant for subsequent phases of donation. Per P4: *"When [the check-in happens] it means that the next information will be a bit deeper and more sensitive than the [most recent] question. So it makes sense."* The researchers nor Ube itself gave any explanation for why check-ins occurred, and P4 had not received any prior check-ins through which they could identify patterns of their occurrence. This informed an important change in check-in placement in Ube–rather than administering these randomly, we placed them where donors already expected them to be: before sensitive groups of questions, in addition to following when donors openly disclosed their sexual experience as unwanted.

5.7 TIDD Phase 6: Retraumatization Management After Donation

TIDD can Realize and Resist the risk of retraumatization (TIA assumptions 1 and 4) after data donation has completed by providing access to trauma and mental health resources regardless of the information gathered about the donor, also also through the opportunity to reflect on and actively confirm that the donor is comfortable providing their data (including the ability to delete of all one's donation). The affordances intent to enact TIA principle #1: Safety even after data donation has officially ended, in recognition that retraumatization may not always be immediate.

5.7.1 Initial ideation with trauma and sexual violence experts. Trauma and sexual violence experts were essential in crafting opportunities for mitigating harm to donors even after donation. In early meetings, they brought up concerns around lasting impacts that may occur. Allowing donors to delete their donation post-participation was intentional so that donors who may have reflected on the experience they donated and self-identified as a victim, or otherwise

have experienced a new stress response related to the donation, may erase their data and associated contextualization responses. To ensure donors understood that this was an option for them, we designed a question at the conclusion of donation asking donors to confirm if they were comfortable providing the data they had provided. The question required donors to either confirm their submission or delete it entirely. This is depicted in Figure 4.

Our experts suggested that since trauma effects may last, it would be essential to have persistent resource availability that donors could return to even well after donation had concluded. NURSE1 advocated for this especially because of the unique, sensitive nature of the sexual experience data. Placement of resources at the conclusion also aligned with TIDD Phase 4: Label-avoidant data contextualization. As UX1 explains, "*I think what we had agreed on is that at the end of data donation, we just across the board for everyone, offer support services, so that we don't imply that a particular person is a victim, based on what they just donated.*"

5.7.2 Ideation with donors. Donors appreciated the inclusion of resources at the conclusion of donation, regardless of whether or not they personally needed them at the time. P17 likened the check-in to an intervention that positively impacted them personally. P17: "*I can answer this question as a survivor of SV [...]. It was literally an intervention that pulled me out of the [bad] situation. So, you know, had these people not reached out to me like I still would be in that situation. So you never know. Even if it just saves one person, if a phone call from a [SV] specialist is going to save that one person, if that's all it does is save one person, you know, then it's a success.*"

However, donors were caught off-guard by the question asking them to confirm their donation, which allowed them to delete their responses. Per P17: "*For the next question. I don't like that [phrasing.] That is planting an idea in my head that I should be concerned for my data like I wasn't worried about it until you said something.*" This runs counter to our intent to adhere to TIA principle #1: Safety because it arguably sparked safety concerns. As a result, we now recommend that donors be provided with resources and options to review, edit, and delete their donation; however, questions or prompts directly asking donors to (re-)consider their donation decision should be avoided so as not to inadvertently imply that the donation was a bad decision.

6 Discussion

In this paper we proposed Trauma-Informed Data Donation (TIDD), an exploratory design framework for donation that acknowledges and designs for the risk of retraumatization when collecting data that may be representative of traumatic experiences. TIDD was created and exemplified through a 2-year process of designing and developing Ube, which supports the donation of data about sexual experiences from online dating. The design effort involved experts on trauma and sexual violence for explicit incorporation of the Trauma-Informed Approach (TIA) from the Public Health domain, as well as data donors who used Ube during observational interviews.

In this section we first reflect on the implications of TIDD to the data donation landscape, including comparisons to other data donation approaches in the literature and opportunity for modifying and combining TIDD with other approaches for context-specific donation of data associated with varying forms of trauma. We then turn out attention to trauma-informed technology design more broadly, particularly to discuss implications of our work on trauma-informed evaluation of technologies and how to involve user-stakeholders in assessment of trauma-informed design.

6.1 Reflections on Applying Trauma-Informed Data Donation

6.1.1 Transferring Trauma-Informed Data Donation to Other Contexts. Trauma-Informed Data Donation was designed, developed, and evaluated in the specific context of online dating sexual violence data, in collaboration with experts on sexual violence and sexual trauma. Because we did not utilize or create TIDD in additional cases we urge readers not to treat TIDD as necessarily generalizable or readily applicable in an as-is state to other data donation contexts and types of trauma. Even if it were, use of TIDD would still require subjective mapping of TIDD's phases to specific interface design choices that would presumably be specific to the particular data donation platform used in each case.

Rather, TIDD would be best described as a malleable, context-specific foundation for incorporating the Trauma-Informed Approach (TIA) into data donation platform design. It is a starting point, or even provocation, for designers to consciously consider how TIA can be foregrounded into data donation processes. To render TIDD transferrable (rather than generalizable), we advise that use of TIDD should necessitate conscious deliberation by future designers of 1) which phases of TIDD apply to their case (and how), and subsequently 2) how to map the deemed-appropriate TIDD phases to interface choices in the chosen data donation platform.

We will use Phase #4 of TIDD (label-avoidant data contextualization) to demonstrate the importance of each of those two decision points. The premise of label-avoidant data contextualization may transfer well to donation of data associated with trauma incurred through person-inflicted harm such as racial or identity-based trauma because these forms of trauma lend themselves to "victim" and "perpetrator" positions and, with that, the risk of implying through careless phrasing that a donor is a victim or perpetrator. However, transferring TIDD Phase #4 to the context of, for instance, natural disaster survivors may not be as useful where labeling is less conducive to adverse affect (i.e., natural disasters have much more objective definitions with universally understood consequences), and may even be beneficial for affirming the donor's experience. Designers for a data donation platform for data associated with a natural disaster may thus opt to omit Phase 4 entirely from their own implementation of TIDD.

Assuming designers have opted to keep Phase #4 (and have exercised similar prudence to other phases), subjective determination is still needed on how each phase should manifest in interface design. We performed this through a two-step mapping process where each TIDD phase was mapped to a discrete interface feature that was also mapped to specific trauma-informed principles it intended to actualize through user engagement with the interface feature. A methodological practice that worked well here was to have experts in sexual trauma drive discussion around labeling risks for the particular trauma, including any known best practices

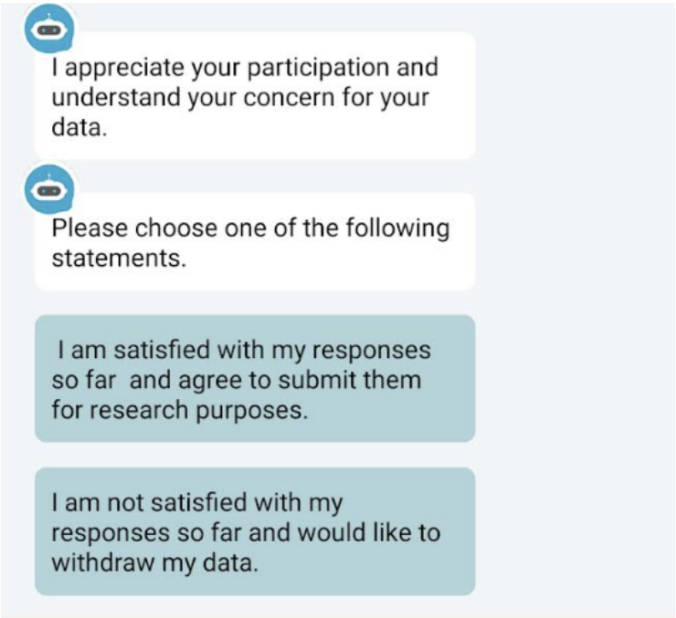


Figure 4: Question designed to allow donors to delete their donation post-participation.

for labeling and, on the contrary, phrases to avoid. It was through this expert-driven crash course on sexual experience labeling that associations with specific trauma-informed principles became clear (in this case, principle 1: Safety and principle 5: Empowerment, voice, and choice). Through multiple interface iterations by the UX members of the team precise phrasings, and placements for those phrasings in our data donation app design, were refined. We would similarly suggest that data donation platform designers allow trauma expert-stakeholders to initiate or drive discussion about the given TIDD phase, with the role of UX designers being to enact the advice through iterative design sketches.

6.1.2 Synthesizing Trauma-Informed Data Donation With Other Data Donation Approaches. TIDD is not at odds with other data donation processes, frameworks, and interface designs; it does not require designers to choose it *instead of* other frameworks. Rather, it is synergistic and augmentative of many prior data donation methodologies. First we demonstrate how TIDD is actually synergistic with three other "holistic" data donation approaches–that is, methodologies that provide steps and/or principles that inform the data donation journey [38] from beginning to end: (1) the "standard" data donation methodology [41], (2) sensitive data donation [41], and (3) participatory data donation [40].

"Standard" data donation methodologies call for basic requirements [41]: donors export a single large file of their personal data from the source platform, provide that file to the data donation platform, and sometimes provide context to their data [41]. However, the details of the process, such as how donors provide contextualization to their data, vary across studies and are not explicitly prescribed; some provide context after donation, while others implement processes to contextualize during donation. TIDD, by comparison, provides a more rigid step-by-step structure for scaffolding donation, and also provides an alternative to the typical "JSON export" style of data donation.

Sensitive data donation (sDD) [41], another holistic methodology for the donation of data, prioritizes involvement of donors as contributors, collaborators, and co-creators, and could be effectively integrated with TIDD. For instance, Principle 2 of sDD (Sensitive Data) encourages platforms to consider how donors can know their sensitive data and draw boundaries around its disclosure. TIDD's Phase 4: Selective data donation addresses this directly by involving donors in the process of creating their data. In fact, co-creation of research goals already satisfies Phase 1 of TIDD. Additionally, incorporating plurality prompts from TIDD can enrich the notion of data donation being dynamic, uniquely situated, and contextualized.

Participatory data donation [40] prioritizes distinguishing between the uploading and curating of data from the official transfer decision; this aligns well with our Phase 4: Selective data donation. The two can provide ideas for one another–perhaps selective data donation in TIDD could happen within the app, which could support JSONs and other large files over excessive manual screenshotting, which doubtless will not work for all data types. Likewise, participatory data donation's emphasis on co-construction of research goals for donated data could enrich Phase 1 of TIDD (equip donors to foresee retraumatization) by allowing donors to shape, rather than just learn about, the dynamics of donation that could be most triggering of trauma.

Outside of holistic data donation approaches, arguably the most common and critical challenge for those designing platforms and methodologies for the donation of data is ensuring informed consent [37, 43, 51, 72, 73, 77, 78, 101, 111, 116]. Prioritization of informed consent in prior work has latent synergy with some TIA principles like "empowerment, voice, and choice," and "collaboration and mutuality" (both of which advocate for acknowledging power

differences), and perhaps most salient, trustworthiness and transparency. While TIDD does not explicitly discuss design for consent, interfaces and frameworks from prior work that support informed consent are complementary and could benefit from TIDD phases such as label-avoidant contextualization (not labeling donors' experiences without their consent) and re-traumatization check-ins (ensuring donors' consent is still valid at the end of the donation journey). Interfaces for informing consent proposed in prior work may be effectively applied and appreciated by TIDD. Gomez Ortega [40] proposes a "framework for meaningfully informed data donation" in which donors are encouraged to reevaluate and reassess their consent to participate as they interact with and learn more about their personal data, which aligns well with TIDD's first phase. Further, value-centered consent interfaces [58, 59, 98], which encourage the foregrounding of a donor's personal values in their consent decisions, could be easily implemented in the TIDD process.

6.2 Reflections on Assessment of Trauma-Informed Design

Incorporation of (or at least inspiration from) the Trauma-Informed Approach (TIA) [48] in technology design has been written about and claimed in various prior works (e.g., [23, 52, 81, 91]). Yet evaluation within HCI of a design artifact's adherence to TIA's four assumptions and six principles remains inconsistent or entirely absent. In this section, we reflect on our methodological decisions for evaluation of Ube's (and TIDD's) adherence to TIA, and future directions for evaluation of trauma-informed design in light of our work.

First we need to re-summarize the state of trauma-informed evaluation in and outside of HCI to establish context for why we had to develop our own evaluation approach. Within HCI, to our knowledge trauma survivor stakeholders have not been involved in evaluating technologies claimed as trauma-informed in their adherence to TIA or impact on retraumatization. Evaluation is not entirely absent though. For instance, Ramjit and colleagues [83] involved advocates from intimate partner violence (IPV) clinics in assessing a new referral system for connecting survivors with services. This involved studying the referral system against context-specific goals (clinic coordination) rather than adherence to specific trauma-informed principles or impact on retraumatization. This limits transferability of their approach to Ube or other design artifacts, but does suggest (perhaps inadvertently) that evaluation of trauma-informed design artifacts should or could be tailored to measures germane to the specific implementation context.

This contrasts with approaches to trauma-informed care evaluation in "traditional" clinical settings in public health, which do utilize standardized scales to assess 1) the perceived adherence of a clinical service to trauma-informed principles according to trauma survivors (recipients of care), and/or 2) TIA knowledge by clinical staff that administer trauma-informed services (for a review, see [79]). These standardized or systematic approaches are tailored to clinical care settings where staff interact with trauma survivors to provide care. They accordingly have limited transferability to Ube and other trauma-informed technologies that are not used for the express purpose of seeking care.

In lieu of directly transferrable evaluation approaches from prior work, our methodology utilized an observational interview protocol that involved donors engaging with the data donation interface and thinking aloud about their donation decisions and reactions to the platform's features. This method required extensive mapping work in the background to connect donor feedback on surface-level interface features to underlying TIA assumptions and principles and, in turn, TIDD phases. Our approach resulted in a sort of "laundered" evaluation of adherence to TIA principles, yet it is the first attempt (to our knowledge) to evaluate a supposedly-trauma-informed technology in its adherence to specific trauma-informed principles. One may argue that this "laundering" would have been unnecessary if Ube users were directly asked for their opinions of individual interface features' adherence to specific trauma-informed principles. We would urge caution here due to a risk of power and knowledge dynamics biasing evaluation. Would users willingly disagree with professionals and "experts" on trauma, and would their (likely hasty) training on TIA not be susceptible to the same biases of those who taught it to them? There is also the risk of labeling, or implying to users that they could or should have experienced trauma, which may not be true especially for technologies not intending to provide care to trauma survivors.

In retrospect, donors certainly did not appear to have a positivity bias: they did not have any issues telling us when they disliked aspects of Ube's design or why they opted to skip certain donation and contextualization prompts. This was beneficial for our team; we were able to improve Ube's platform design based on donor experiences. We would thus advocate that future trauma-informed computing efforts consider a similar TIA mapping approach that tethers precise trauma-informed principles to precise interface features, and then solicits user feedback on those features as conduits or proxies for TIA adherence.

Perhaps the most glaring limitation of our approach to evaluating Ube is the absence of retraumatization measurement. We did not assess if TIDD reduces retraumatization, or if it does so better than other data donation approaches. Yet this problem is not unique to our case. While TIA is predicated on resisting retraumatization, we are not aware of evaluation approaches, measures, or scales (inside or outside of HCI) that directly measure retraumatization or its mitigation by trauma-informed services or artifacts. This is almost certainly not because of failure to consider retraumatization as an aspect worthy of evaluating. Ramjit and colleagues mention "ethical limitations" that prevent controlled, comparative trials of trauma-informed artifacts [83], which we would agree with. Another ethical challenge with assessing *re*traumatization is the necessity of initial trauma. This can be challenging to assess in trauma-informed technologies that are not necessarily intended "for" trauma survivors, as is our case with Ube. Having users fill out retraumatization surveys or scales can itself be labeling and incur its own negative psychological response.

Taking a cue from Ramjit and colleagues [83], future work in trauma-informed computing could assess technologies not through explicit retraumatization measures, but with context-specific measures that may serve as proxies of retraumatization. For instance, our own team can continue to collect feedback during large-scale deployment Ube through, for example: (1) tracking failures to complete data donation or decisions to delete donations, (2) reviewing

retraumatization check-in responses, and/or (3) monitoring access to mental health resource documentation.

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Table 2: Demographic information about participants for observational interviews.

Donor	Age	Reported sexual harm through dating apps	Self-reported Gender and Sex	Race or Ethnicity	State
P1	23	Yes	Gender Non-Conforming	White or Caucasian	Michigan
P2	27	Yes	Man	Black or African-American	Michigan
P3	26	Yes	Woman	Black or African-American	Michigan
P4	30	Yes	Woman	Black or African-American	Michigan
P5	29	Yes	Man	Black or African-American	Pennsylvania
P6	25	No	Woman	Middle Eastern or North African	Michigan
P7	30	No	Man	White or Caucasian	Michigan
P8	25	Yes	Man	White or Caucasian	Washington
P9	44	No	Cisgender Woman	Latino, Hispanic, or Spanish origin	Florida
P10	32	Yes	Transgender Man	Black or African-American	New york
P11	19	Yes	Non-Binary	White or Caucasian	Michigan
P12	32	Yes	Cisgender Woman	Black or African- American	California
P13	25	No	Cisgender Man	White or Caucasian	California
P14	22	No	Cisgender Woman	Asian	Michigan
P15	21	No	Cisgender Woman	Black or African-American	Michigan
P16	22	No	Cisgender Woman	Asian	California
P17	42	No	Cisgender Woman	White or Caucasian	Michigan
P18	21	No	Cisgender Woman	White and Black	Michigan

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A Demographic information